

Practical #5

Practical Title:

To understand the structure of a basic web application and its role in a DevOps pipeline

Theory:

1. What is a Web Application?

A web application is a software application that runs on a web server and is accessed through a web browser (like Chrome).

Examples: login forms, student portals, feedback systems.

2. Components of a Web Application

Frontend:

- User interface (UI)
 - Built using **HTML, CSS, JavaScript**
 - Runs in the browser
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Backend

- Handles logic and processing
 - Technologies: Node.js, Java, Python
 - Communicates with database
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Database

- Stores application data
 - Examples: MySQL, MongoDB
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3. What is DevOps Pipeline?

A DevOps pipeline is an automated process that allows developers to **build, test, and deploy applications continuously**.

It includes stages like:

- Code → Build → Test → Deploy → Monitor

Procedure:

Step 1: Create a Basic Web Page (HTML)

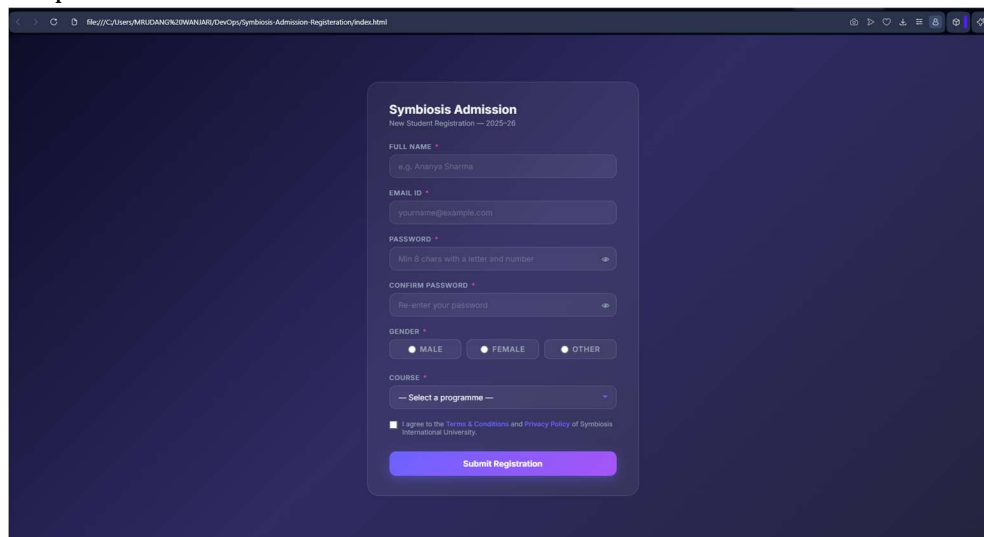
Step 2: Add Styling using CSS

Step 3: Add Simple JavaScript Functionality

Step 4: Run the Web Page in Browser

Step 5: (Optional) Upload the Web App to GitHub

Output Screenshots:



The screenshot shows a web browser window with a registration form titled "Symbiosis Admission". The form is set against a dark blue background. The form fields include:

- FULL NAME ***: A text input field with a placeholder "e.g. Ananya Sharma".
- EMAIL ID ***: A text input field with a placeholder "yourname@example.com".
- PASSWORD ***: A text input field with a placeholder "Min 8 chars with a letter and number".
- CONFIRM PASSWORD ***: A text input field with a placeholder "Re-enter your password".
- GENDER ***: Three radio button options: **MALE**, **FEMALE**, and **OTHER**.
- COURSE ***: A dropdown menu with the text "Select a programme".
- Agreement**: A checkbox labeled "I agree to the Terms & Conditions and Privacy Policy of Symbiosis International University."
- Submit Registration**: A prominent purple button at the bottom of the form.

Activity / Task:

Task 1: Simple Web Application

Description:

A simple **Student Admission Form** where users enter their name and submit the form. It uses HTML for structure, CSS for styling, and JavaScript for interaction.

Task 2: Use in DevOps Pipeline

- Code stored in GitHub
- Automatically built and tested
- Deployed to server using CI/CD
- Monitored for performance

Flow:

Developer → GitHub → CI/CD Tool → Deployment → Monitoring

Task 3: Steps to Deploy Application

1. Create server (local/cloud)
 2. Install web server (Apache/Nginx)
 3. Upload files to server
 4. Configure server
 5. Access via browser (URL)
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Advantages

1. Easy to develop and deploy
 2. Platform independent
 3. Supports automation (DevOps)
 4. Scalable
 5. Accessible from anywhere
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Challenges

1. Security issues
2. Browser compatibility
3. Requires backend for complex logic